

Lab 2: GO CONCURRENCY

Net-Centric Programming

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Part 1: Your First Goroutine

1.1 Three Counter

```
o thanhphatchau@Ants-MacBook-Pro lab2 % go run .  
  
=== Go Concurrency Lab Menu ===  
1. Three Counters  
0. Exit  
Choose option: 1  
  
=== Part 1 – Task 1.1: Three Counters ===  
Counter A: 1  
Counter B: 1  
Counter C: 1  
Counter C: 2  
Counter B: 2  
Counter A: 2  
Counter A: 3  
Counter C: 3  
Counter B: 3  
Counter B: 4  
Counter A finished!  
Counter C: 4  
Counter C: 5  
Counter B finished!  
Counter C finished!  
All done!  
  
=== Go Concurrency Lab Menu ===  
1. Three Counters  
0. Exit  
Choose option: 1
```

1.2 Website fetcher

```
○ thanhphatchau@Ants-MacBook-Pro lab2 % go run .
```

```
=== Go Concurrency Lab Menu ===
```

- 1. Three Counters
- 2. Website fetcher
- 0. Exit

```
Choose option: 2
```

```
=== Part 1 - Task 1.2: Website Fetcher ===
```

```
Fetching Google.com...
```

```
Fetching Twitter.com...
```

```
Fetching Facebook.com...
```

```
Fetching Amazon.com...
```

```
✓ Got data from Twitter.com
```

```
✓ Got data from Google.com
```

```
✓ Got data from Amazon.com
```

```
✓ Got data from Facebook.com
```

```
Completed in: 501.381125ms
```

Part 2: Channels

2.1 Calculator with channels

```
=== Go Concurrency Lab Menu ===
```

- 1. Three Counters
- 2. Website fetcher
- 3. Calculator with Channels
- 0. Exit

```
Choose option: 3
```

```
=== Part 2 - Task 2.1: Calculator with Channels ===
```

```
Numbers: [10 20 30 40 50]
```

```
Sum: 150
```

```
Average: 30.0
```

```
Time: 505.669291ms
```

2.2 Message Queue

```
bye.  
thanhphatchau@Ants-MacBook-Pro lab2 % go run .
```

```
=== Go Concurrency Lab Menu ===
```

1. Three Counters
2. Website fetcher
3. Calculator with Channels
4. Message Queue
0. Exit

```
Choose option: 4
```

```
=== Part 2 - Task 2.2: Message Queue ===
```

```
Message 1 from Charlie
```

```
Message 1 from Bob
```

```
Message 1 from Alice
```

```
Message 2 from Alice
```

```
Message 2 from Bob
```

```
Message 2 from Charlie
```

```
Message 3 from Charlie
```

```
Message 3 from Alice
```

```
Message 4 from Charlie
```

```
All messages received!
```

Part 3: WaitGroup

3.1 File downloader

```
○ thanhphatchau@Ants-MacBook-Pro lab2 % go run .
```

```
=== Go Concurrency Lab Menu ===
```

1. Three Counters
2. Website fetcher
3. Calculator with Channels
4. Message Queue
5. File Downloader
0. Exit

```
Choose option: 5
```

```
=== Part 3 - Task 3.1: File Downloader ===
```

```
Downloading photo.jpg (2MB)...
```

```
Downloading video.mp4 (8MB)...
```

```
Downloading song.mp3 (4MB)...
```

```
Downloading doc.pdf (5MB)...
```

```
Downloading archive.zip (6MB)...
```

```
✓ photo.jpg complete!
```

```
✓ song.mp3 complete!
```

```
✓ doc.pdf complete!
```

```
✓ archive.zip complete!
```

```
✓ video.mp4 complete!
```

```
✓ All downloads complete! (801.452375ms)
```

3.2 Number processor

```
○ thanhphatchau@Ants-MacBook-Pro lab2 % go run .
```

```
=== Go Concurrency Lab Menu ===
```

1. Three Counters
2. Website fetcher
3. Calculator with Channels
4. Message Queue
5. File Downloader
6. Number Processor
0. Exit

Choose option: 6

```
=== Part 3 - Task 3.2: Number Processor ===
```

Numbers: [1 2 3 4 5 6 7 8 9 10]

Evens: [2 4 6 8 10]

Odds: [1 3 5 7 9]

Squares: [1 4 9 16 25 36 49 64 81 100]

Time: 510.969208ms

Part 4: Select

4.1 Search Rate

```
○ thanhphatchau@Ants-MacBook-Pro lab2 % go run .
```

```
=== Go Concurrency Lab Menu ===
```

1. Three Counters
2. Website fetcher
3. Calculator with Channels
4. Message Queue
5. File Downloader
6. Number Processor
7. Search Race
0. Exit

Choose option: 7

```
=== Part 4 - Task 4.1: Search Race ===
```

Engine B won! (201.724542ms)

Results from Engine B for 'golang concurrency'

Part 5: Real – World Application

5.1 Library simulation

```
thanhphatchau@Ants-MacBook-Pro lab2 % go run .
```

```
Student 39 left library after 2 hours
Student 98 entered library, will study for 1 hours
Student 51 left library after 3 hours
Student 97 entered library, will study for 4 hours
Student 46 left library after 4 hours
Student 72 entered library, will study for 2 hours
Student 67 left library after 3 hours
Student 76 entered library, will study for 2 hours
Student 79 left library after 3 hours
Student 90 entered library, will study for 2 hours
Student 48 left library after 3 hours
Student 99 entered library, will study for 1 hours
Student 43 entered library, will study for 4 hours
Student 52 left library after 1 hours
Student 89 entered library, will study for 1 hours
Student 55 left library after 4 hours
Student 88 entered library, will study for 2 hours
Student 87 entered library, will study for 2 hours
Student 98 left library after 1 hours
Student 71 left library after 1 hours
Student 82 left library after 2 hours
Student 91 entered library, will study for 2 hours
Student 57 left library after 3 hours
Student 99 left library after 1 hours
Student 94 left library after 2 hours
Student 85 left library after 2 hours
Student 89 left library after 1 hours
Student 42 left library after 2 hours
Student 80 left library after 4 hours
Student 60 left library after 4 hours
Student 36 left library after 4 hours
Student 81 left library after 4 hours
Student 78 left library after 4 hours
Student 77 left library after 2 hours
Student 75 left library after 2 hours
Student 72 left library after 2 hours
Student 53 left library after 3 hours
Student 76 left library after 2 hours
Student 93 left library after 3 hours
Student 90 left library after 2 hours
Student 56 left library after 4 hours
Student 88 left library after 2 hours
Student 54 left library after 3 hours
Student 91 left library after 2 hours
Student 92 left library after 4 hours
Student 87 left library after 2 hours
Student 41 left library after 3 hours
Student 43 left library after 4 hours
Student 74 left library after 4 hours
Student 97 left library after 4 hours
Student 95 left library after 4 hours
```

```
=== Simulation Complete ===
```

```
Total students served: 100
```

```
Library was open for: 10 hours
```